

Alexander Robinson

Embedded Systems & IoT Firmware Engineer • 12 Years Experience • MERIQ Index: 90% (Mastered)

■ alexander.robinson686@example.com | ■ +1 (908) 916-3906 | ■ Tokyo, Japan

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: I2C/SPI/UART, FreeRTOS, ARM Cortex, C/C++, Python, PCB Debugging, MQTT

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Tesla* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core I2C/SPI/UART service, reducing latency by 32%.
- Led a squad of 8 engineers delivering scalable microservices processing 22M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 51%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2014 - 2023)

- Designed and maintained FreeRTOS data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using ARM Cortex saving \$36k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, FreeRTOS, ARM Cortex, C/C++]

Engineered end-to-end production solution leveraging I2C/SPI/UART and FreeRTOS with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [ARM Cortex, C/C++, Python, PCB Debugging]

Architected high-resilience system integrating ARM Cortex with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Electrical and Computer Engineering

Massachusetts Institute of Technology (MIT) — 2010 - 2014 (Score: 3.46 / 4.0)